

GWAS–PheWAS-Based Strategy for Utilizing Genomic and Phenotypic Data in Peach (*Prunus persica* (L.) Batsch)

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STUDY OVERVIEW

A reusable Genome-Wide Association Study (GWAS)–Phenome-Wide Association Study (PheWAS) framework continuously integrates newly accumulated phenotypic data with a shared genomic dataset in peach germplasm. GWAS identifies candidate SNPs associated with each newly evaluated trait, while PheWAS re-examines these variants across multiple accumulated phenotypes to identify cross-trait associations. This strategy enables repeated trait discovery and supports marker development and digital breeding applications.

1 Peach germplasm



Whole-Genome Resequencing (WGRS)
(NovaSeq 6000)

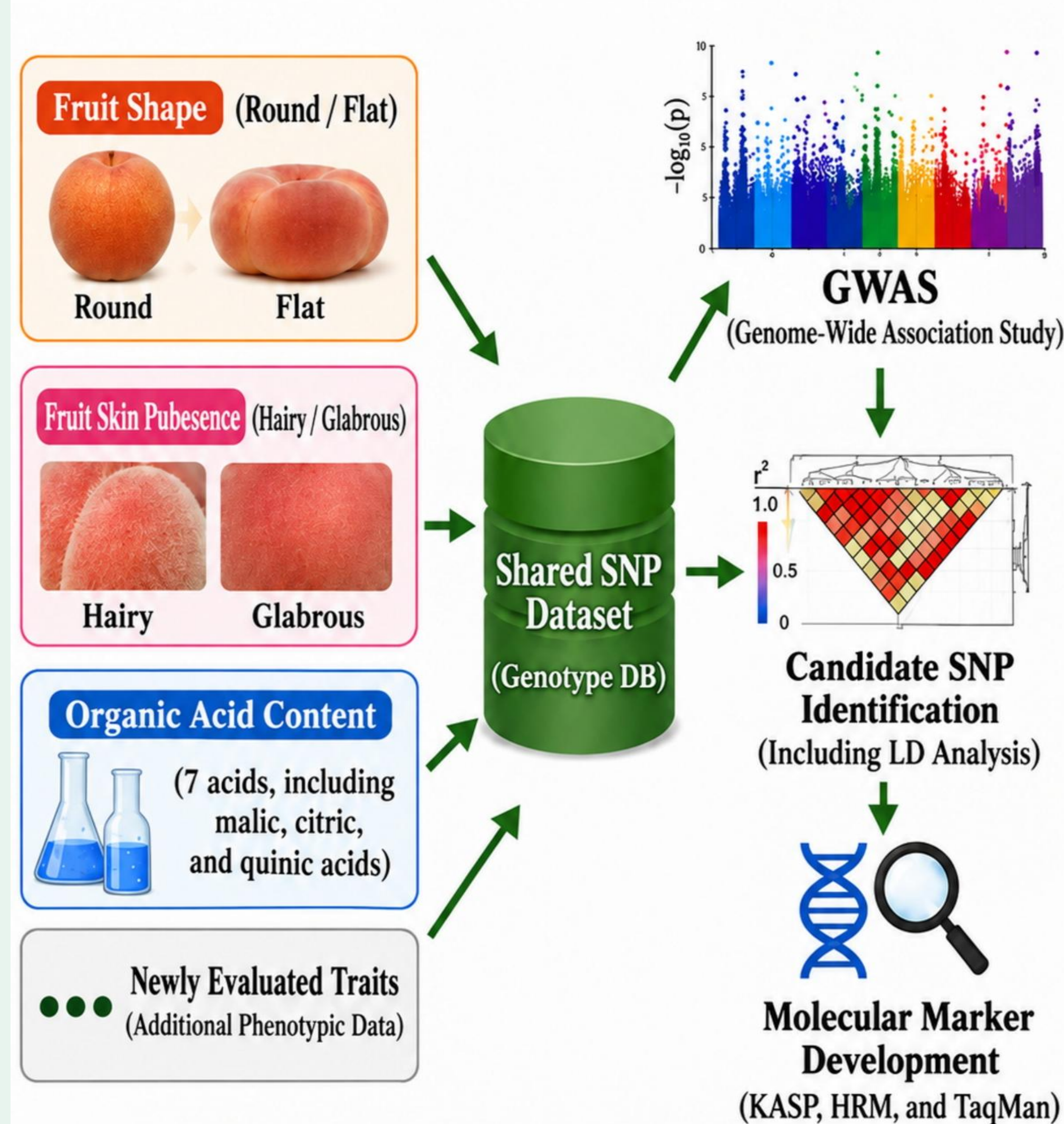
Reference Genome Alignment and Variant Calling

High-Quality SNP Dataset
(~0.9 Million SNPs)

- ✓ Population Structure Analysis (ADMIXTURE, PCA)
- ✓ Core Collection Selection

Shared Genomic Information (SNP Data)

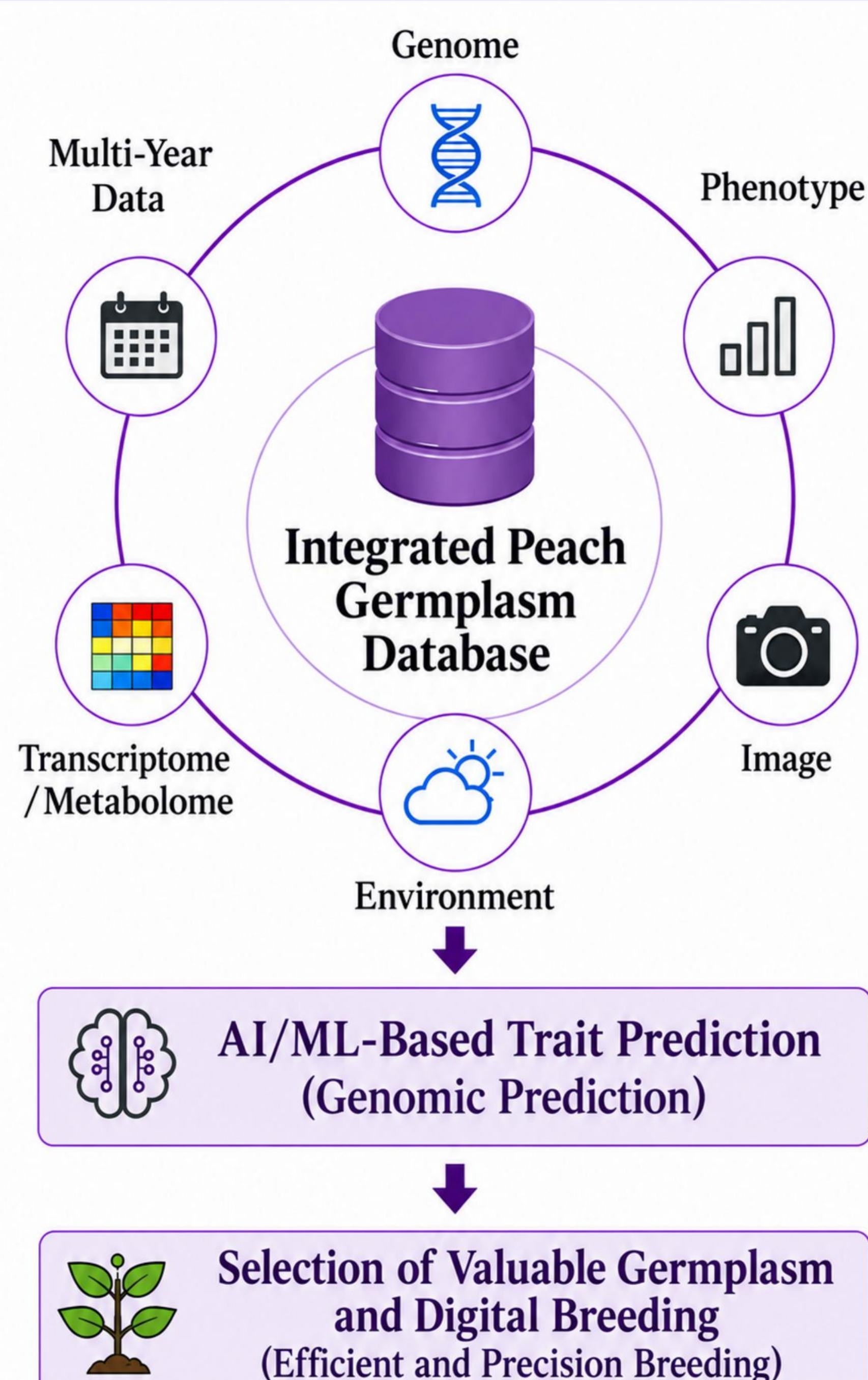
2 Continuous integration of phenotypic data



Each newly collected phenotypic dataset is linked to the existing genomic resource for repeated analysis and application.

Build the genotype dataset once and continuously expand the phenotypic data.

3 Multi-layer data integration



A Reusable Framework for Peach Digital Breeding

The same genomic backbone is leveraged repeatedly as new phenotypic data are added to enable continuous trait discovery.

